

WELD DEFECT INSPECTION REPORT

AI-Powered Visual Inspection System

Report ID	RPT-2026-00002
Sample ID	1213
Scan Number	#2
Scan Date & Time	11 May 2026, 15:55
Operator	pranay
Location / Site	Welding Bay 3, Plant A — Pune
Specimen Type	Plate / Flat
Joint Configuration	Double V Butt Joint
Material	Stainless Steel
Specimen Size	300 x 500 mm
Welding Process	SMAW (Shielded Metal Arc Welding)
Industry / Application	Power Generation
Detection Device	Inspection Camera

1. SCAN SUMMARY

The AI-powered visual inspection of sample **1213** (Scan #2) was performed on **11 May 2026, 15:55** using the AI model (version v1 (Simulation)). The scan covered the **CAP** side of the weld on a **Plate / Flat** specimen.

1	1	CRITICAL	74.4%
Total Defects	Defect Types	Highest Severity	

Defects Detected:

#	Defect Type	Count	RPN	Severity	Surface Visible	Accuraccy
1	Crack	1	400	CRITICAL	Yes	74.4%

2. DETAILED DEFECT FINDINGS

2.1 Crack	CRITICAL	RPN: 400
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Description:
Linear discontinuity caused by rapid cooling, thermal shrinkage, or base material incompatibility. Can propagate under cyclic loading and cause catastrophic failure.

Detection Details:

Instances Detected:	1
Accuracy:	74.4%
Surface Visible:	Yes

Primary Causes:
Rapid cooling, high residual stress, hydrogen embrittlement, incompatible filler material, improper preheat.



Applicable Standards:
Rejected per ISO 5817 Level B; AWS D1.1 rejects all cracks; ASME BPVC Section VIII - no cracks permitted.

Recommended Action:
Immediate repair required. Grind out defective area and re-weld. Perform supplementary NDT (UT/RT) to verify full removal before re-welding.